

## Experiment 13 :

Generate a parse tree for a given sentence using a context-free grammar using python program.

### Code:

```
import nltk

from nltk import CFG

from nltk.parse import ChartParser

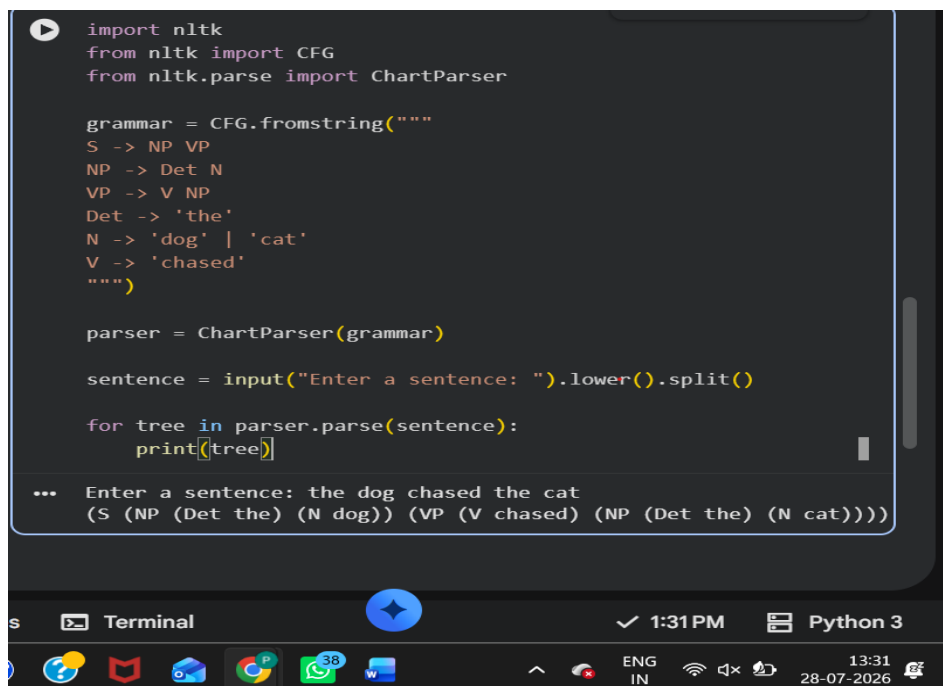
grammar = CFG.fromstring("""
S -> NP VP
NP -> Det N
VP -> V NP
Det -> 'the'
N -> 'dog' | 'cat'
V -> 'chased'
""")

parser = ChartParser(grammar)

sentence = input("Enter a sentence: ").lower().split()

for tree in parser.parse(sentence):
    print(tree)
```

### Output



```
import nltk
from nltk import CFG
from nltk.parse import ChartParser

grammar = CFG.fromstring("""
S -> NP VP
NP -> Det N
VP -> V NP
Det -> 'the'
N -> 'dog' | 'cat'
V -> 'chased'
""")

parser = ChartParser(grammar)

sentence = input("Enter a sentence: ").lower().split()

for tree in parser.parse(sentence):
    print(tree)
```

... Enter a sentence: the dog chased the cat  
(S (NP (Det the) (N dog)) (VP (V chased) (NP (Det the) (N cat))))